QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

UMHackathon 2026

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Document Control

Field	Detail
System Under Test (SUT)	*UMHackathon 2026 – Business Optimizer*
Team Repo URL	*https://github.com/wongyiien-creator/UMHackathon_TeamHehe.git*
Project Board URL	*https://trello.com/invite/b/69e70611d24dec4a6fe64caf/ATTIeb59aedb51189be034c9832e7312ae16072B8586/business-optimizer*

Objective:

The primary objective is to ensure that Business Optimizer can handle the features off the app, login, sign up, market analysis, market generation, compare marketing mix, getting context through Vertex AI Search, getting data from Firestore ,save data in Firestore and organized display of information

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

This section aligns the testing connected back to specific user requirements via Requirement Traceability Matrix". So, it ensures every test has been conducted to prevent bugs/failure in products for that specific requirements and unplanned feature creep.

1.1 In-Scope Core Features

- **Market Analysis and Marketing Mix Generation:** Get saved data, Z.ai processing and display data in UI
- **Login:** Input username and password, verify credentials in firestore
- **Sign Up:** Input username and password, verify credentials and save into firestore
- **Marketing Mix Comparison:** Input details, get context in Vertex AI Search, Z.ai processing and display data in UI
- **Save Data:** Input data name and content, save into datastore if content not over limit

1.2 Out-of-Scope

- Profile Management: Edit profile info, view previous result
- An emergency rollback feature and golden version
- Automatically saving comparison, generation and analysis into firebase

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- Fallback to older model once current Z.ai model limit has been reached

2. Risk Assessment & Mitigation Strategy

Quality Assurance risks are anticipatory. So, identify the technical risk associated with Business Optimizer's requirements, architecture. The risk is to be evaluated using the 5*5 Risk assessment Matrix, Risk Score = Likelihood * Severity.

Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
AI Hallucination: *Z.ai generates marketing plans that are logically flawed*	*3*	*5*	*15 (High)*	*Check quantitative result of the marketing mix generation to ensure the number is not abnormally large. If number is too large, user will be asked to try again.*	*Input a prompt that forces high quantitative numbers. Verify the UI displays an retry message instead of displaying the result*
Empty Fields: *No input is given but the system is allowed to continue*	*1*	*4*	*4 (Low)*	*Check each input field is field before moving on using .isEmpty(). If a field is found empty, User is asked to field up the field before the system can continue.*	*Leave input field empty when entering. Verify that the system stays on the page and inform the user about the empty field*
Context Window Overflow: *Too much data sent to Z.ai, causing the end of the data to be ignored.*	*3*	*3*	*9 (Medium)*	*Before saving into Firestore, frontend check if the content is over the word count set. If it is, it will be blocked*	*Paste a text that is exactly 1 word over the limit. Verify the an error message pops up instead of successfully saving*
Outdated Strategy: *Outdated marketing mix strategy given due to old market data*	*3*	*4*	*12 (Medium)*	*Ensure system prioritize getting data that have been uploaded by the user at most 30 days ago. If no data is found, only the system will get the latest 10 dataset uploaded by*	*Manually edit Firestore timestamps to make some data 40 days old. Run the app and check to ensure only data within 30-day-old data was used.*

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Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
				the user regardless of date	

Risk Assessment Scoring Criteria:

Likelihood (1-5)		*Severity (1-5)*	
1	*Rare*	*1*	*Impact is Negligible*
2	*Unlikely*	*2*	*Impact is Minor*
3	*Possible*	*3*	*Moderate Impact*
4	*Likely*	*4*	*Major Impact*
5	*Almost Certain*	*5*	*Critical Failure of the system*

Risk Score = Likelihood × Severity

Risk Level Reference:

Risk Score	***Risk Level***	***Recommended Action***
1 – 5	*Low*	*Monitor only. Acceptable risks.*
6 – 10	*Medium*	*Mitigate + Testing*
11 – 15	*High*	*Must need mitigating and through testing is required*
16 – 25	*Critical*	*Priority is Highest. Need extensive level of testing.*

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3. Test Environment & Execution Strategy

The testing of Business Optimizer will take place, how we are going to handle test data and the rules/threshold that we are incorporating to govern your testing phases.

- **Unit Test**
  - *Scope: Input check, AI Output Parser, Word Limit Filter*
  - *Execution: Written using Kotlin test and PyTest framework. These run locally in Android Studio or in CLI.*
  - *Isolation: To focus only on the input and output check, Firestore and Z.ai are mocked.*
  - *Pass Condition: 100% success rate on "Happy Cases" (no empty fields, AI not hallucinating, within word limit) and 100% unsuccessful rate on"Negative Cases"*
- **Integration Test**
  - *Scope: The connection between the frontend inputs, Vertex AI Search (context retrieval), and the Z.ai Agent (generation logic). The bridge between the input validation logic and Firebase Firestore storage.*
  - *Execution: Performed after proper completion of Unit Testing. Ensuring the input checker successfully passes data to Z.ai and Vertex AI Search only when fields are complete. It also verifies that the word limit filter correctly prevents large payloads from being sent to or stored in Firebase. It validates that the Z.ai response passes through the AI output parser before being displayed to the user.*
  - *Workflow: To ensure the API calls and interactions with databases are working, so real calls are tested without mocking.*
  - *Pass Condition: Marketing Mix is generated using real-time search data and passes the AI output parser. Comparison of Marketing Mix is only generated when no empty fields are found. Data is only saved if it satisfies the word limit filter.*
- **Test Environment (CI/CD Practice):**
  - *Local Testing: This testing is manual one on localhost*
  - *Staging/CI: GitHub Actions triggers tests on every push of code.*
  - *CI/CD automated pipeline: So, GitHub Actions does basic tests for branch specific level and then PyTest for PR (Pull requests) to Main branch*
- **Regression Testing & Pass/Fail Rules:**
  - *Execution Phase: This testing tests across all in-scope features each time a new branch is merged with the main branch.*
  - *Pass/Fail Condition: Test case pass is only considered when actual outcome perfectly aligns with expected one. If mismatch/threshold not achieved, the test status is considered fail and need to incorporate that in Defect Log*
  - ***Continuation Rule:** While doing testing in a gradual manner, the initial cases must need to meet the threshold to allow continuation for further testing*
- **Test Data Strategy:**
  - *Manual: Created Mock Users account: New Account (no data uploaded) and Old Account (few data uploaded)*
  - *Automated: Using JSON seed scripts to populate the Firebase Emulator. This allows the system to be tested with hundreds of data.*
- **Passing Rate Threshold:**

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- *A minimum of 90% of all test cases that are executed must pass. For the critical tests, all the tests (100%) need to be passed and there is no exception for that. Because critical failure costs huge in production*

4. CI/CD Release Thresholds & Automation Gates

This section is mainly for defining metrics & threshold for success criteria. So, if a certain threshold is not met, the pipeline blocks the forward transition.

4.1 Integration Thresholds (Merging to Main)

The primary needs of this threshold to be crucial is because the main branch must be a stable source of truth. So, it needs to be achieved through continuous integration checks.

Checks	***Requirements***	***Project***	***Pass/Failed***
Automatic Build	***Zero Compilation Errors (Kotlin & Python)***	***No Errors***	***Pass***
Unit Tests	***100% Success on Input & Parsing logic***	***100% success rate***	***Pass***
Code Quality	***Zero code error***	***Few warnings, no error***	***Pass***
Test Coverage	***Minimum 81.2%***	***70%***	***Failed***

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4.2 Deployment Thresholds (Pushing to Production)

Checks	***Requirements***	***Project***	***Pass/Failed***
Regression Test	***100% Success in Login & Save functions***	***100% success rate***	***Pass***
AI Output Rate	***Min 90% of output must be complete and not hallucinated***	***83.33% pass rate***	***Failed***
Critical Bugs	***Zero Fatal Exception***	***Zero Fatal Exception***	***Pass***
API Performance	***Vertex AI Search Response < 3s***	***41.89s***	***Failed***
Security	***No API key expose***	***Z.ai API key hidden***	***Pass***

5. Test Case Specifications (Drafts)

This section is for you to draft my test cases. The test cases include:

- ***1 Happy Case (Entire Flow)***
- ***1 Specific Case (Negative/Edge Test)***
- **2 Non-Functional (NFR) Tests (Performance/Security & Privacy)**

Test Case ID	**Test Type & Mapped Feature**	**Test Description**	**Test Steps**	**Expected Result**	**Actual Result**
TC-01	***Happy Case (Entire Flow):*** *Marketing Mix Generation*	*Verify that a logged-in user can successfully input data, have Z.ai analyze and come up with a Marketing Mix Strategy. Ensure the*	*1. Open app, login with credentials.* *2. Upload a few market and business data*	*Uploaded data passes the limit test and Z.ai generates a relevant plan based on user*	*Data is successfully stored in Firestore. Plan generated in 49.70s. All fields*

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Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
		strategy is not hallucinated	*(10 dataset)* *3. Click Analysis button* *4. Review Analysis and strategy*	*uploaded data. The output passes the sanity check.*	*populated.* *Status: Passed*
TC-02	***Specific Case (Negative): Word Limit***	*Verify that the system blocks data that exceeds the character limit to prevent database bloat and AI errors.*	*1. Go to Save data page* *2. Paste a dataset exceeding the set character limit* *3. Attempt to click "UPLOAD"*	*The system should block the input and display an error message saying it exceeded word limit. The data will no be saved into Firestore*	*Input is blocked, displaying an error message. Data not saved into Firestore* *Status: Passed*
TC-03	***NFR (Performance): AI Generation Latency***	*Ensure that the Z.ai processing and Vertex AI Search retrieval do not exceed the acceptable wait time for a user.*	*1.Upload more data (3 unstructure dataset)* *2. Measure time from "Click" to "UI Display."* *3. Repeat 3 times to find the average.*	*Average response time should be < 10,000ms (10 seconds) to ensure a smooth user experience.*	*Average Response time is 36.76s* *Status: Failed*
TC-04	***NFR (Security & Privacy): Data Isolation (Multi-tenancy)***	*Verify that a User cannot access, data belonging to another User ID during market analysis*	*1. Log in as User A and save a data* *2. Log out and log in as User B* *3.Click start analysis*	*Analysis will only use User B's uploaded data for market analysis*	*Only User B's data is used. User A's data remains inaccessible for User B.* *Status: Passed*

6. AI Output & Boundary Testing (Drafts)

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This section is designed to validate that your GLM integration does produce acceptable output and handles any inputs that are abnormal gracefully.

6.1. **Prompt/Response Test Pairs**

Document at least 3 Prompts/response test pairs. For each of them, do define the acceptance criteria - what a correct, acceptable or a failing response may look like for your use case.

Test ID	Prompt Input	Expected Output (Acceptance Criteria)	Actual Output	Status
AI-01	**System Prompt + Data A:** (Unstructured data only: Social media trend and reviews)	Output should suggest a marketing angle based on viral trends	The AI identified a "Viral Growth Trap" where aesthetic TikTok trends are driving demand. It also focuses on the other comments, shifting from cheap import to premium brand and automating WhatsApp commerce.	Pass
AI-02	**System Prompt + Data B:** (Structured data only: Inventory levels and sales spreadsheets for 3 months)	Output must identify the "Best Seller" and suggest a optimization strategy based on the structured data	The AI recommended a hybrid athleisure footwear captures the 50% market preference for sports shoes and sneakers. Implementing free delivery thresholds to boost average order value	Pass
AI-03	**System Prompt + Data A & Data B:**	The output should be more detailed	The AI suggested a strategic shift	Pass

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	(Unstructured data and Structured data).	with more context-aware reasoning	toward B2B wholesale and agri-tech branding to capitalize on new logistics hub interests and the padi price increase. It recommended prioritizing an emergency fertilizer procurement and weatherproofing storage o mitigate a critical inventory imbalance before expanding into the Shah Alam market.	

6.2. Oversized/Larger Input Test

In this section, do define the maximum input size your system accepts. Do highlight here what happens when the limit is exceeded.

Fields	Details
Maximum Input Size	10,000 Character Counts
Input used while testing	10,001 Character Counts
Expected Behavior	Reject
Actual Behavior	Reject
Status	Pass

6.3. Adversarial/Edge Prompt Test

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Data Uploaded by user is ambiguous such as "the stuff is expensive” but not specifying what is the “stuff”. GLM will continue using the data and lead to strategy about the wrong product

Status: Fail

6.4. Hallucinating Handling

Once Z.ai produce a response, it will first go through an AI output parser, where it make sure the forecasted revenue growth percentage is not abnormally big. If it pass, the full response will be displayed on the UI. If it fails, the response will be blocked and user must try again.

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